MILITARY SPECIFICATION

CARTRIDGE, 105MM, HE, ROCKET ASSISTED, XM548, ROCKET PROPELLANT GRAIN FOR

1. SCOPE

1.1 This specification contains requirements not normally required by the drawings and provides quality assurance provisions for fabrication of the rocket propellant grain for the 105MM Rocket Assisted Cartridge, XM548.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or requests for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons General Specification For.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-453 - Inspection Radiographic.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1315

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DRAWINGS

ARMY

9212412 - Grain, Propellant, Rocket, 105MM (Inhibited).
9227372 - Carton, Packing, Ammunition for Grain, Propellant, Rocket, 105MM (Inhibited).
9227373 - Box, Packing, Ammunition for Grain, Propellant, Rocket, 105MM (Inhibited).

PUBLICATIONS

ARMY

ET-9212412 - Equipment Tabulation.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Carpet Roll (see 6.4). - The carpet roll used in extruding the propellant grain shall comply with Specification MIL-P-60464.

3.3 Propellant Grain.-The inhibited propellant grain shall comply with all requirements specified on drawing 9212412 and with all requirements specified in applicable specifications.

3.4 Cracks, fissures and foreign material.-The propellant grain shall contain no crack or fissure measuring more than 0.30 inch in any direction and no foreign material measuring more than 0.20 inch in any direction. The total combined length of all cracks or fissures for the two halves (one grain) shall not exceed 0.60 inches and the combined length of all foreign material (maximum dimensions of each particle of foreign matter) shall not exceed 0.40 inch. Cracks, fissures and foreign material measuring less than 0.031 inch (considering largest dimensions) shall be disregarded.

3.5 Duds.-The grain in the test motor shall function on the first attempt after the boron potassium nitrate pellets have ignited.

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3.6 Inhibitor separation.-The propellant grain shall contain no individual separations between the inhibitor and the grain greater than 0.060 square inches and the total area of separation for the two halves shall not exceed 0.25 square inches.

3.7 Static functioning. - The slope of the normal pressure time trace after the initial ignition pressure peak shall be progressively negative and shall contain no pressure peak more than 300 psi above the normal operating curve for a duration of more than 0.030 seconds.

3.7.1 Maximum pressure

3.7.1.1 Minus 65°F. - The average of the maximum pressures shall be no less than 1220 psi and shall be no greater than 1330 psi. The standard deviation for group shall be no greater than 46.4 psi. No individual maximum pressure shall be less than 1115 psi and no greater than 1435 psi.

3.7.1.2 Plus 145°F. - The average of the maximum pressures shall be no less than 1365 psi and shall be no greater than 1475 psi. The standard deviation for the group shall be no greater than 103 psi. No individual maximum pressure shall be less than 1040 psi and no greater than 1800 psi.

3.8 Process.-In the process of extruding the propellant grain, a 4 support spider shall be used in the approach section of the extrusion die.

3.9 Workmanship.-The propellant grain shall be free of dirt, grease, chips, gouges, inhibitor solvent drippings (Elba), and other defects which would make the item unsuitable for its intended use.

3.10 First Article Inspection. - This specification makes provisions for first article inspection. Requriements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detailed specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals, all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 First Article Inspection

4.2.1 Submission. - The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 22 propellant grains which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures, and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used

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in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings/ certificates of conformance for materials, (see 4.1.1.1.e). A Sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. - Propellant grains will be subjected by the Government to any orall of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawing. After examination for other requirements, twenty propellant grains shall be static fired as specified in 4.4.3, ten at each of the applicable temperatures. Two propellant grains shall be tested for compressive strength as specified in 4.4.2.

4.2.3 Rejection. - If any grain fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection Provisions

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous quantity of propellant grains from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of propellant grains from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of propellant grains produced by one manufacturer, in accordance with the same specifications, and same specification revisions, same drawings and same drawing revisions. Each lot shall consist of propellant grains which

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have been subjected to the same chemical and physical processes intended to make the final product homogeneous. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL-STD-1168. Unless otherwise approved by the contracting officer, the inspection lot size of propellant grains deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Each lot of propellant grains shall contain not more than one lot of carpet rolls from one manufacturer.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Once agreement has been reached as to which method will be used (group or individual) that method will be used for all of the classification of defects listed in this specification. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Grain, propellant, prior to applying inhibitor (see dwg. 9212412).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical: None defined.			
Major:	AQL 0.40 percent		
	101. Outside diameter.....	Gage	01001
Minor: None defined.			

4.3.2.2 Grain, propellant, prior to spliting (see dwg. 9212412).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 1.00 percent		
	101. Outside diameter of inhibited grain.....	Gage	02001

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102.	Outside diameter at .31 basic length	Gage	02002
103.	Overall length	Gage	02003
104.	Inhibitor separation (See 3.6)	Visual	02004
105.	Boron potassium Nitrate pellet flush with or below end surface	Visual	02005
106.	Excess adhesive on grain end surface	Visual	02006
107.	Depth of 2 holes in end surface max	Gage	02007
108.	Perpendicularity of tapered end surface	Gage	20008

Minor: AQL 1.50 percent

201.	Inside diameter of grain, min	Gage	02009
202.	Concentricity of inside diameter to outside diameter of inhibitor	Gage	02010
203.	Evidence of poor workmanship (see 3.4)	Visual	02011

4.3.2.3 Grain, Propellant, after splitting (see dwg. 9212412).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Finished weight	Balance	03001
Minor:	AQL 2.50 percent		
201.	Split symetrical with outside diameter of inhibitor	Gage	03002
202.	Surface finish improper	Visual	03003
203.	Marking missing, misleading or illegible	Visual	03004
204.	Evidence of poor workmanship (See 3.9)	Visual	03005

4.3.3 Testing

4.3.3.1 Cracks, fissures and foreign material (see 3.7).- Major defect - Code No. 03006.-This test shall be performed 100 percent on each half of every grain. The test shall be conducted in accordance with 4.4.1. Any grain containing defects in excess of the requirements of 3.4 shall be classed defective and removed from the lot.

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4.3.3.2 Static Testing

4.3.3.2.1 Minus 65°F - Major defect - Code No. 03007.-A Sample of ten (10) propellant grains shall be selected at random from each lot and tested in accordance with 4.4.3. If any grains fail to meet the requirements of 3.5, 3.7, or 3.7.1, the lot shall be rejected.

4.3.3.2.2 Plus 145°F - Major defect - Code No. 03008.-A sample of 10 propellant grains shall be randomly selected from each lot and tested in accordance with 4.4.3. If the grains fail to meet the requirements of 3.5, 3.7, or 3.7.1, the lot shall be rejected.

4.3.3.3 Compression test (for informational purposes only).

4.3.3.3.1 Minus 40°F - Code No. 03009.-Three samples shall be selected from each lot and conditioned at minus 40°F for 12 hours, minimum. The compressive strength shall be recorded for each specimen. The test shall be conducted as specified in 4.4.2 using test equipment as specified in 4.3.4.

4.3.3.3.2 Plus 140°F - Code No. 03010.-Three samples shall be selected from each lot and conditioned at plus 140°F for 12 hours minimum. The compressive strength shall be recorded for each specimen. The test shall be conducted as specified in 4.4.2 using test equipment as specified in 4.3.4.

4.3.3.4 Width of split - Major defect - Code No. 03011. - The width of the cut taken in splitting the grain shall not exceed the drawing requirement. For verification the width of cut shall be inspected at the beginning of each 8 hours of production by making a partial cut of the grain, retracting the saw blade and measuring the width of cut.

4.3.4 Inspection Equipment - Equipment Tabulation Number ET-9212412 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection equipment. The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test Methods and Procedures

4.4.1 Radiographic examination of propellant grains.-The split propellant grain shall be radiographed with the flat side against radiographic film and the longitudinal axis of the grain perpendicular to the radiographic beam. Observations shall be made for cracks, fissures and foreign material. Any unit containing one or more defects in excess of size or total lengths permitted shall be classed defective. In the event that cracks, fissures, or foreign material cannot be clearly defined by one radiograph, the grain shall be placed on end and a second radiograph taken with the radiographic beam parallel to the longitudinal axis. Radiographic inspection shall be in accordance with MIL-STD-453. The penetrameter required in MIL-STD-453 shall be designed and constructed so as to assure a 2 percent minimum sensitivity.

4.4.2 Compression strength.-A specimen 2 inches long and 1 inch in diameter shall be prepared from each sample selected under 4.2.3.3 and conditioned at the required temperature for 12 hours, minimum. The specimen shall be placed in the required test equipment and compressed at a loading rate of 2 inches per minute.

4.4.3 Static test.-The propellant grain shall be loaded into the test motor. The test motor with the boron potassium nitrate pellets and delay assembly in place shall be conditioned at the required temperature for 12 hours, minimum. The test motor shall be removed from the conditioning chamber, assembled to the required fixture and fired within 3 minutes of removal from the conditioning chamber. Pressure, time and thrust shall be recorded for each test.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C - Preservation and packaging of grains shall be in accordance with dwg. 9227372.

5.2 Packing

5.2.1 Level C - Packing of grains shall be in accordance with dwg. 9227373.

5.3 Marking - Marking shall be in accordance with applicable drawings.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. - The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists. - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.2 Inspection equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.5 Submission of contractor designs. - All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND4. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Carpet roll. - In the event the carpet roll has been manufactured and stored more than six months, a sample shall be selected from each carpet roll lot and tested in accordance with MIL-P-60464. The test shall be conducted not more than 30 days prior to extruding carpet rolls into propellant grains.

6.5 Submission of first article sample. - Instructions as to the location for the evaluation of the first article sample shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NI.

Custodian:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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